

The general solution and the first two derivatives for a beam undergoing bending vibration are

$$Y(x) = A \sin(\beta x) + B \cos(\beta x) + C \sinh(\beta x) + D \cosh(\beta x), \quad (1)$$

$$\frac{dY(x)}{dx} = \beta(A \cos(\beta x) - B \sin(\beta x) + C \cosh(\beta x) + D \sinh(\beta x)), \quad (2)$$

$$\frac{d^2Y(x)}{dx^2} = \beta^2(-A \sin(\beta x) - B \cos(\beta x) + C \sinh(\beta x) + D \cosh(\beta x)). \quad (3)$$

The boundary conditions for the simply supported beam are

$$Y(x)|_{x=0} = 0, \quad (4)$$

$$M(x)|_{x=0} = EI \frac{d^2Y(x)}{dx^2} \Big|_{x=0} = 0, \quad (5)$$

$$Y(x)|_{x=L} = 0, \quad (6)$$

$$M(x)|_{x=L} = EI \frac{d^2Y(x)}{dx^2} \Big|_{x=L} = 0. \quad (7)$$

From Eq. (4) and Eq. (5) we get

$$Y(0) = A \sin(0) + B \cos(0) + C \sinh(0) + D \cosh(0) = 0 \rightarrow D = -B, \quad (8)$$

$$M(0) = \beta^2(-A \sin(0) - B \cos(0) + C \sinh(0) - B \cosh(0)) = -2\beta^2 B = 0 \rightarrow B = 0. \quad (9)$$

Entering the values of B and D in Eq. 1 yields

$$Y(x) = A \sin(\beta x) + C \sinh(\beta x). \quad (10)$$

Applying Eq. (7) to Eq. (10) yields

$$\frac{d^2Y(x)}{dx^2} \Big|_{x=L} = \beta^2(-A \sin(\beta L) + C \sinh(\beta L)) = 0 \rightarrow C = A \frac{\sin(\beta L)}{\sinh(\beta L)}, \quad (11)$$

and finally

$$Y(x)|_{x=L} = A \sin(\beta L) + A \frac{\sin(\beta L)}{\sinh(\beta L)} \sinh(\beta L) = 2A \sin(\beta L) = 0. \quad (12)$$

From this we obtain

$$\sin(\beta L) = 0, \quad (13)$$

with the solutions

$$s_n = \beta_n L = n\pi, \quad n = 1, 2, 3, \dots \quad (14)$$